

Data Science Fundamentals in R
Introduction to transparent and reproducible data science
Fall 2026 — In Person Delivery/Online
OCN682 / MBIO612 • Slack Channel OCN682

Lecture: Tuesday 1:00 – 2:50 pm, BIL 335

Flip class: Any time before start of class, at home (first half of semester)

<i>Instructor:</i> Dr. Nyssa Silbiger e-mail: silbiger@hawaii.edu phone: 808-956-0327 Office Hours: Tuesday 3:00 – 4:00 Office: BIL 335/MSB 612	<i>Teaching Assistant:</i> Mica Chapuis e-mail: mchapuis@hawaii.edu Office Hours: Monday 12:00 – 14:00 Office: MSB 306
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Free Online Required Readings

Wickham and Grolemund: R for Data Science — r4ds.hadley.nz

Wilke: Fundamentals of Data Visualization — clauswilke.com/dataviz

Healy: Data Visualization: A Practical Introduction — socviz.co

Mastering Shiny — mastering-shiny.org

Quarto — quarto.org

Tidy Modeling with R — tmwr.org

Papers: Browman and Woo (2018); Lowndes et al. (2017); Jennings et al. (2023)

Supplemental Texts (optional, but helpful): Wickham, Advanced R — adv-r.hadley.nz

Course Objectives

This course is intended to make you think like a coder and provide you with a general understanding of how to conduct transparent and reproducible data science. This course is NOT a statistics or bioinformatics course. You will learn how to script and use tools to make your data examination more transparent. The specific learning objectives include:

- Be able to share your code and data on a public version-controlled repository (GitHub).
- Create a reproducible script.
- Work collaboratively on a project.
- Learn efficient and effective ways to explore your data.
- Create high-quality visuals.
- Learn to love working with data!
- Develop the professional judgment to use generative AI tools ethically, transparently, and critically as one part of a broader reproducible-research toolkit — without letting AI substitute for learning the underlying fundamentals.

Course Description

This course combines a series of lectures with hands-on computer labs and group work. The goal is for you to be an active participant in lectures because doing something is the best way to learn it. Further, you will also learn by teaching; therefore, you will have multiple opportunities for peer learning and teaching in this course. Your grade for the class will be based on coding assignments, presentations, short quizzes, a community data project, and a final project (see Grading, below). There are no exams in this class; however, learning a coding language, like any language, requires lots of practice and repetition to master it. Therefore, this class will be a lot of work both in and outside of the classroom.

In lectures, you will learn fundamental concepts in how to think like a coder and create reproducible data. You are encouraged and expected to ask questions during lectures. Given the short time in class for a 3-unit course, you will also have lectures at home for the first half of the semester. These can be completed at any time but **MUST BE DONE BEFORE THE NEXT CLASS**. At home, you will work on coding homework assignments. All textbooks are free and available online; you are expected to complete all required readings before class.

New this semester: this course has been redesigned with support from UH Mānoa's Integrating Experiential Learning (EL) initiative. In Week 8, students will complete an AI-Assisted Community Data Sprint, in which you will act as a coding consultant for a real local community partner (e.g., DAR, DOH, or a Mālama restoration group), using a generative AI tool as a coding assistant under a documented, disclosed, and critically evaluated workflow. This component is layered on top of — not a substitute for — the core coding skills you build independently throughout the rest of the semester. See the Ethical & Responsible AI Use Policy and the AI-Assisted Community Data Sprint sections below for full details.

Grading

Component	Percent of Grade
Short Quizzes	15%
Homework Assignments	25%
AI-Assisted Community Data Sprint & Reflection	20%
Class Participation	5%
Final Independent Project	35%
TOTAL	100%

Your grade will be based on the components listed above.

Short quizzes are intended to encourage you to keep up with the readings and gain an understanding of the vocabulary associated with coding. These are maximum 10-minute, multiple-choice, open-note quizzes that occur at the beginning of each class. Do not come late — there will be no extra time allowed. **I will drop your lowest grade, so there are no make-up quizzes.**

Weekly homework assignments will be turned in independently, but you are allowed to have them checked by your classmates first. All assignments must be turned in on GitHub for full credit (this will be explained in class). Assignments will be graded according to the rubrics posted each week on the

class website. Unless an assignment explicitly says otherwise (see the Ethical & Responsible AI Use Policy), these assignments are meant to build your own hands-on coding fluency and **must be completed without generative AI assistance. If I suspect AI use in your homework you will be asked to live code the assignment in class.**

AI-Assisted Community Data Sprint & Reflection. In Week 8, you will be paired with a real coding roadblock from a local community partner (e.g., DAR, DOH, or a Mālama group). Working in small teams, you will use a generative AI tool (e.g., Claude, ChatGPT, Gemini, Posit Assist) as a coding assistant to debug the partner's code, write the necessary functions, and produce a clean, reproducible example (Reprex) that solves the problem while keeping a structured AI Developer Log documenting your prompts, the AI's output and errors, how you diagnosed and corrected those errors, and a short reflection on how the experience shaped your confidence and professional identity as an emerging data scientist.

Class Participation will be based on your preparedness (i.e., did you complete the readings and assignments before class), answering and asking good questions, showing up in person, helping your fellow classmates, and participating in think, pair, shares. This is a collaborative course, and participating is critical to your learning.

Final independent project and presentation. You will use the skills you learned throughout the semester to tell a story with your own data (or publicly available data) and present it to the class according to the rubric on the class website. As with weekly homework, the final project is meant to demonstrate your own coding fluency and is subject to the same restrictions as weekly homework assignments.

Your grade will be based on your percentage of the total possible points. You can use the following cutoffs as a guideline. I reserve the right to lower the grade cutoffs, but do not count on that happening.

Percent	Letter Grade
93-100	A
90-92.9	A-
88-89.9	B+
83-87.9	B
80-82.9	B-
78-79.9	C+
73-77.9	C
70-72.9	C-
68-69.9	D+
60-67.9	D
<59	F

Ethical & Responsible AI Use Policy

Generative AI tools (e.g., Claude, ChatGPT, Gemini, GitHub Copilot) are powerful, increasingly standard parts of a professional data scientist's toolkit — and they are also genuinely useful for building

career-ready skills. At the same time, this course exists to make you a confident, independent coder: the ability to write, debug, and **reason through your own R code is the whole point, and it cannot be outsourced to AI while you are still building it.** This policy sets out two clearly different modes for when and how AI tools may be used in this course.

1. Core skill-building work (quizzes, weekly homework, and the final project). The purpose of these assignments is for you to build your own coding fluency. For this work:

- Do not use generative AI to write, complete, or generate the code, functions, or written interpretations you submit.
- You may use google, stack overflow, or other non-AI sources to look up general syntax, understand an error message, or clarify a concept as long as you are the one writing and understanding the logic of your own solution.
- Submitting AI-generated code or AI-generated written work as your own, without disclosure, **is a violation of this course's academic integrity policy** (see Cheating and Plagiarism, below) and will be treated accordingly.
- **If I suspect your code was written with AI for an assignment, I will ask you to live code your answers in front of the class and you will be given a grade of 0 for that assignment.**
- If you are unsure whether a particular use of AI is appropriate for a specific assignment, ask before you submit.

2. The AI-Assisted Community Data Sprint (Week 8). This assignment explicitly requires and teaches responsible AI-assisted coding, once you already have the independent skills to evaluate what AI produces. For this assignment:

- You are required to use a generative AI tool as a coding assistant, and to critically evaluate, correct, and document its output rather than accept it uncritically.
- Every prompt, AI response, error, and correction relevant to your solution must be logged in your AI Developer Log.
- You remain fully responsible for the correctness, functionality, and reproducibility of the final code you deliver to the community partner. "The AI wrote it" is never an acceptable explanation for broken or incorrect code.

3. Disclosure. For any assignment where you use AI in a way this policy permits, include a brief AI-use statement (what tool you used, what you asked it, and what you used or changed) at the top of your submission or in your commit message. Transparency about AI use is itself a graded professional competency in this course.

4. Data and privacy. Never paste a community partner's proprietary, sensitive, or personally identifiable data into a public or unvetted AI tool. Use only the sanitized/example data provided for the Data Sprint unless I tell you otherwise.

5. Why does this policy exist? Employers increasingly expect both independent technical competence and AI literacy. Introducing AI too early, or without guardrails, undermines the coding fluency this course exists to build; introducing it not at all leaves you unprepared for professional practice. This policy is designed to give you both a strong independent foundation first, then a single, well-scaffolded, documented experience using AI the way a working data scientist would.

Questions about what is or is not permitted for a specific assignment should be directed to Dr. Silbiger or the TA before the assignment is due.

Attendance

This is an intellectually challenging and time-demanding course. Attendance is important to be successful. I understand that we may have field trips or conferences during the semester for our research. Please let me know about dates you will need to miss for your research-related activities.

Communication

Outside of class time, I will communicate with you via your UH email account, Lamakū, and our dedicated Slack channel. If you do not normally use your UH email account, please configure it to forward your messages to your preferred account. You can expect me to respond to your email or Slack messages within 24 hours (except on weekends or if I am in the field). The Slack channel will be used for you to easily communicate with your classmates and to ask for advice and share resources. [Use this link to sign into our class channel and follow the directions.](#)

Your best source of course information is the course GitHub site (github.com/OCN-682-UH/Fall_2026) and Lamakū. Check the schedule on the website regularly because occasionally we may have to change the schedule due to unforeseen complications.

Expectations

You can expect me to be punctual, treat you fairly, return your graded assignments in a timely fashion, and, most importantly, treat you with respect. In return, I expect you to treat other students, the TA, and me with respect and do everything you can to ensure a comfortable learning environment. For example, jokes or comments made in poor taste that make any member of the class uncomfortable will not be tolerated. Use of cell phones, tablets, or computers during class for any purpose other than those directly related to the class (e.g., note-taking and coding) is not allowed.

Diversity and Equity

It is my goal to create a learning environment that supports diversity of thoughts, perspectives, and experiences, and honors your identities. To help create an inclusive environment:

- Please share your pronouns at the beginning of the semester with me.
- If you feel like your performance in the class is being impacted by your experiences outside of class, please do not hesitate to come and talk with me. I want to be a resource for you.
- I am continuing to learn about diverse perspectives and identities. If something was said in class (by anyone, including me) that made you feel uncomfortable, please talk to me about it (anonymous feedback is always an option).

Policies

Cheating will result in failure of the course and may result in further University discipline. The UH policy states: “Cheating includes, but is not limited to, giving unauthorized help during an examination, obtaining unauthorized information about an examination before it is administered, using inappropriate sources of information during an examination, altering the record of any grade, altering an answer after an examination has been submitted, falsifying any official UH Mānoa record, and misrepresenting the facts in order to obtain exemptions from course requirements.”

Plagiarism will result in you failing the assignment. Plagiarism is “Plagiarism includes, but is not limited to, submitting, to satisfy an academic requirement, any document that has been copied in whole or in part from another individual's work without identifying that individual; neglecting to identify as a quotation a documented idea that has not been assimilated into the student's language and style; paraphrasing a passage so closely that the reader is misled as to the source; submitting the same written or oral material in more than one course without obtaining authorization from the instructors involved;

and ‘dry-labbing,’ which includes obtaining and using experimental data from other students without the express consent of the instructor, utilizing experimental data and laboratory write-ups from other sections of the course or from previous terms, and fabricating data to fit the expected results.” Don't do it. Undisclosed or out-of-policy use of generative AI (see Ethical & Responsible AI Use Policy, above) is treated as a form of cheating/plagiarism under this policy. If you are uncertain about what constitutes plagiarism or acceptable AI use, talk to me.

Late assignments lose 10% per day, i.e., the most you can receive for an assignment that is one day late is 90% of its point value, two days late 80%, etc.

Punctuality: You must be on time for all classes.

Students with disabilities must register with the KOKUA Program, QLCSS 013, phone (808) 956-7511 (Voice/TTY) or (808) 956-7612 (Voice/Text), email kokua@hawaii.edu.

Students and faculty each have responsibility for maintaining a safe and respectful space to express their reasoned opinions. Professional courtesy and consideration for our classroom community are especially important with respect to topics dealing with differences in race, color, gender and gender identity/expression, sexual orientation, national origin, religion, disability, and age.

Title IX

As your instructor, one of my responsibilities is to help maintain a safe learning environment on our campus. In the event that you choose to write, speak, or otherwise disclose having experienced sexual misconduct/sexual violence — including rape, sexual assault, sexual battery, dating violence, domestic violence, or stalking — and specify that this occurred while you or the perpetrator were a UH student, federal and state laws require that I, as a “responsible employee,” notify our campus Title IX Coordinator. The Title IX Coordinator will contact you to inform you of your rights and options as a survivor and connect you with support resources, including possibilities for holding accountable the person who harmed you. Know that you will not be forced to share information, and your level of involvement will be your choice.

UH Title IX Coordinator: Jennifer Rose, Hawaii Hall, Room 112. Phone: (808) 956-2299. E-mail: t9uhm@hawaii.edu, manoa.hawaii.edu/titleix.

If you do not want the Title IX Coordinator notified, instead of disclosing the experience to me, you can speak confidentially with our Campus Care Advocate.

UH Confidential Advocate: Leslie Cabingabang, UH Confidential Advocacy, email advocate@hawaii.edu.